Date
27/7/26

CO2-AT-2

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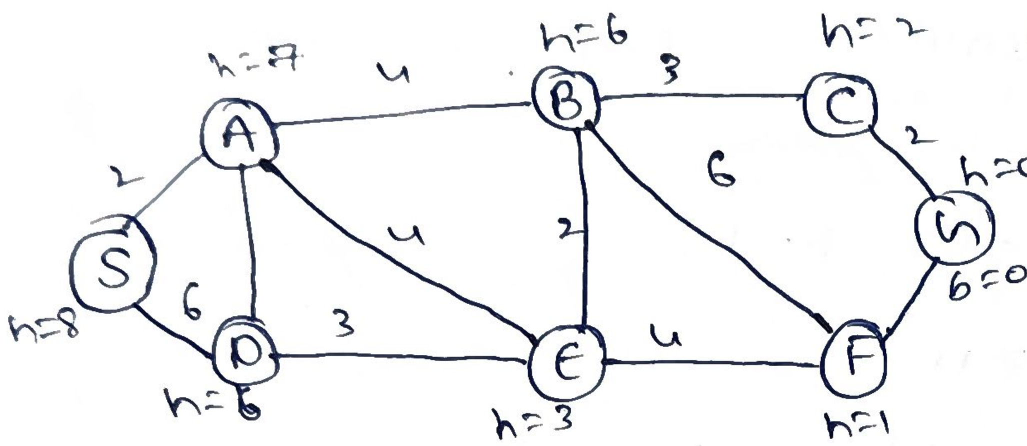
MLA-0101 (A-I)

Q,)



92 A6

Q) Smart Ambriance Navigation: 5 total 10

$f(n) = g(n) + h(n)$

Node	g(n)	h(n)	f(n) = g+h
S	0	8	8
A	2	7	9
B	6	6	12
D	8	6	14
E	6	3	9
F	10	1	11
G	12	0	12

* S→A:

g(A) = 2, h(A) = 7

$f(A) = 2+7 = 9.$

* Expand A $\Rightarrow$ B: $g = 2+4 = 6$, $f = 6+6 = 12$
  D: $g = 2+6 = 8$, $f = 8+6 = 14$
  E: $g = 2+4 = 6$, $f = 6+3 = 9$.

Q2)

* choose E because $f=9$,

* From E:

  - F: $g = 6+4 = 10$, $h = 1$

  $F(S) = 10+1 = 11$

  $g(G) = 10+2 = 12$

Selected route

$S \to A \to E \to F \to G$

Total cost

- $2+4+4+2 = 12$

- Another Route

  $S \to A \to B \to C \to G$

$\to$ cost $2+4+3+2 = 11$

$12 > 11$

$S \to A \to B \to C \to G$, with cost 11.

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[illegible]

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

[illegible]

[illegible]

, one positive [illegible]

[illegible]

So, [illegible]

* Replacing

(4,5) → (4,6) → (3,6)

At (3,6), [illegible]

[illegible]

[illegible] (3,6) → (2,5) → [illegible]

Q2) Autonomous warehouse Robot :

S	?	?	?	?	?	?
?	#	?	?	#	?	?
?	?	?	#	?	?	#
?	#	?	?	?	#	?
?	?	#	?	?	?	G

G = (1, 7) S = (5, 1)

$h(n) = |r - r_g| + |c - c_g|$

- one possible intial path is

(5,1) → (5,2) → (5,3) → (5,4)

→ (5,5) → (5,6) → (5,7)

So, (5,7) → (4,7)

* Replanning

(4,7) → (4,6) → (3,6)

At (3,6), it discovers another obstacle

(2,6)

Now, (3,6) → (3,5) → (3,4)

Then (3,4) → (2,4) → (1,4)

Finally (1,4) → (1,5) → (1,6) → (1,7)

- one valid onile execution

S → (5,2) → (5,3) → (5,4) → (5,5) → G3)

(5,6) → (5,7) → (4,7)

Replan :-

(3,6) → (3,5) → (3,4) → (2,4) → (1,4)

Then :

(1,4) → (1,5) → (1,6) → G

onile	vs offilne search
• obstacles are init-ally unknown	• complete map is known
• learn while moving	• plans before moving
• Replanning is Required	• usually no Replanning
• may tare extra steps	• usually more efficient

∴ The roboot should use onile A*.

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Q) University examination Scheduling.

A) CSP Formulation

$X = \{ E_1, E_2, E_3, E_4, E_5, E_6 \}$

$D(E_1) = D(E_2) = \ldots D \ldots = D(E_6)$

$= \{ T_1, T_2, T_3, T_4 \}$

Exam	Students	Possible Halls
E_1	100	A
E_2	80	A, B
E_3	70	A, B
E_4	60	A, B, C
E_5	90	A
E_6	50	A, B, C

Hall Capacities

A = 120

B = 90

C = 60

B) constraints

$E_1 \neq E_2$, $E_1 \neq E_3$, $E_2 \neq E_4$, $E_3 \neq E_5$,

$E_4 \neq E_6$, $E_2 \neq E_5$.

c) MRV + Forword checking.

$E_2 = T_1$

exam	Remaining domain
E_1	$\{T_2, T_3, T_4\}$
E_3	$\{T_1, T_2, T_3, T_4\}$
E_4	$\{T_2, T_3, T_4\}$
E_5	$\{T_2, T_3, T_4\}$
E_6	$\{T_1, T_2, T_3, T_4\}$

Time Slot	Examination	Hall	Students
T_1	E_2	B	80
T_1	E_3	A	70
T_1	E_6	C	50
T_2	E_1	A	100
T_2	E_4	C	60
T_3	E_5	A	90
T_4	-	-	-

verification

$E_1 \rightarrow E_2$ - T_2 vs T_1 ✓

$E_1 \rightarrow E_3$ - T_2 vs T_1 ✓

$E_2 \rightarrow E_4 \rightarrow T_2$ vs T_2

$E_3 \rightarrow E_5 \rightarrow T_1$ vs T_3

$E_2 \rightarrow E_6 \rightarrow T_2$ vs T_1 ✓

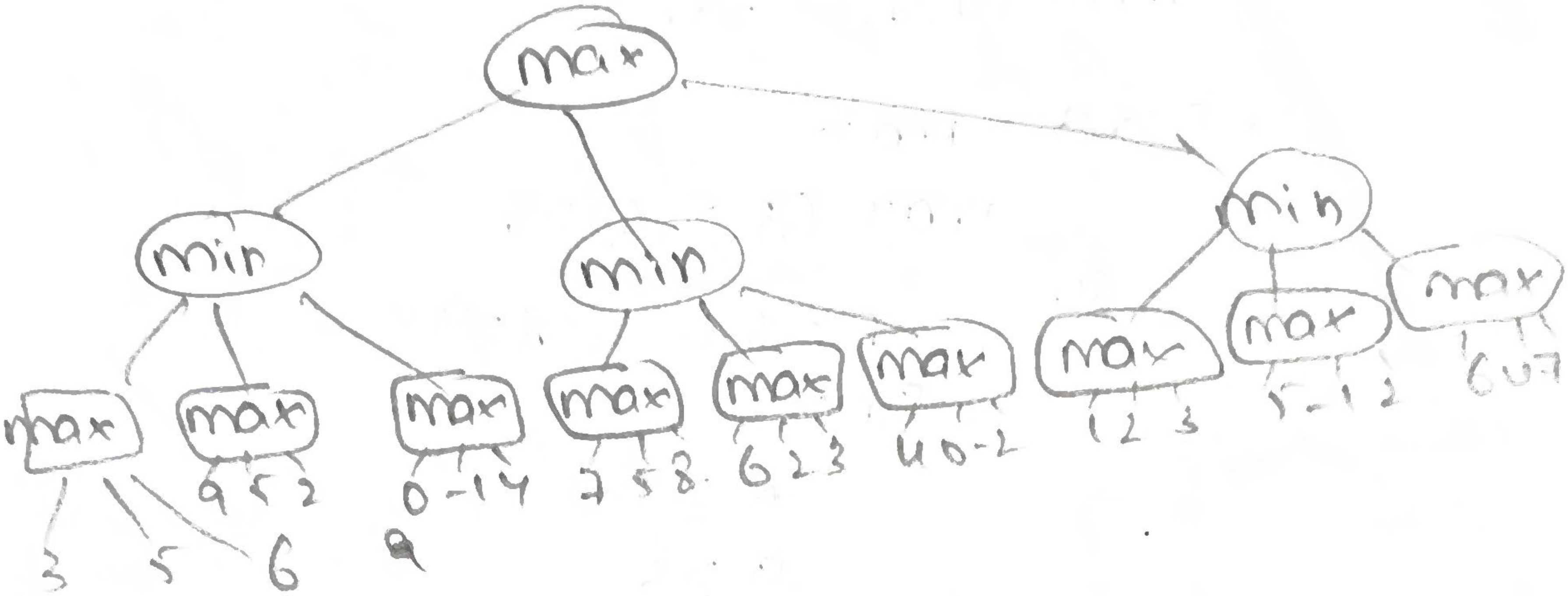
$E_2 \rightarrow E_5 \rightarrow T_1$ vs T_3

Final Answer

$E_1 = T_2$, $E_2 = T_1$, $E_3 = T_1$, $E_4 = T_2$,

$E_5 = T_3$

Game - playing AI.



Three max nodes contain.

[3, 5, 6] , [9, 1, 2], [0, -1, 4]

max = 6   max = 9   max = 4

min chooses

min = 4   ∴ left = 4

. middle Min Branch

[7 5, 8] [6, 2, 3] [4, 0, -2]

max values = 8, 6, 4

min (8, 6, 4) = 4.

→ middle = 4.

Right min Branch.

(1, 2, 3) (5, -1, 2) [6, 4, 7]

max values = 3, 5, 7

min (3, 5, 7) = 3

∴ Right = 3.

Root values = 4, 4, 3

max (4, 4, 3) = 4.

- First max

max (3, 5, 6) = 6

so $\beta = 6$, $\alpha = 9$

Since

$\alpha \geq \beta$

$9 \geq 6$

max (0, -1, 4) = 4

Middle Branch

[7, 5, 8] → 8

[6, 2, 3] → 6

[4, 0, -2] → 4

→ At min node:

$\beta \leq \alpha$ $3 \leq 4$

Final Alpha-Beta Result.

minimax value = 4.

- Entire [5, -1, 2] Branch
- Entire [6, 4, 7] Branch
- 1, 2 from the secound max.

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Problem Analysis

	D	A	B	C	E
D	-	6	9	8	7
A	6	-	3	5	6
B	9	3	-	4	5
C	8	5	4	-	3
E	7	6	5	3	-

Solution	Route	Cost
S_1	D → A → C → B → E → D	26 km
S_2	D → B → A → C → E → D	30 km
S_3	D → A → B → E → C → D	24 km
S_4	D → E → C → B → A → D	27 km
S_5	D → A → E → B → C → D	23 km
S_6	D → C → B → A → E → D	29 km

- First Hill-climbing move

compare

26, 30, 24, 27, 23, 29

23km

∴ S_5 = D → A → E → B → C → D.

S_5 - Best neighbour = 22km.

* move from Sr.

∴ 23 → 22

since all neighbours of this new solution ≥ 22.

• Final Result According to the Assignment.

* Final local cost = 22km.

• Cost of every neighbouring solution ≥ 22

* optimization strategy:-
Hill climbing /local search

→ Initial best route → S5 = 23 km

Improved route → 22km

• Final local -search cost = 22km

Reason:- local search considers only neighbouring solutions.